

Compositional Linearization for Superposition and Hamiltonianization

Maximilien Péroux

David I. Spivak

Abstract

test

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1 Introduction

Doing stuff is important. ♦David says: No it's not. ♦ ♦Maxi says: Yes it is. ♦

2 Monoidal Promonads

2.1 Definition and Quadratic Forms

2.2 General Theory of Quadratic Forms

2.3 Algebras for Monoidal Promonads

2.4 Moore Internalization for Promonadic Lenses

2.5 Lenses for Monoidal Promonads

3 Parametrized Systems and Linearization

3.1 A Reminder on Para

3.2 The Monoidal Product on Lin

3.3 Monoidal Retractions Between Lin and Crit

3.4 Lifting to Coalgebras over Poly

4 Applications and Comparisons

4.1 Superposition and Physics

4.2 The Poincaré Point of View

4.3 Linearization of Differential Equations